Does Financial News Sentiment Move the S&P 500?

DATA 498D Capstone Project Proposal

Problem Statement

A common assumption in finance is that public information is reflected in prices almost immediately. If markets process information efficiently, then news headlines should offer little reliable advantage for predicting short-term returns. However, this idea is often challenged in practice. Professional investors use expensive news terminals, quantitative firms build text-based trading signals, and retail traders frequently react to headlines in real time. These behaviors suggest that market responses to news may be more gradual or inconsistent than theory implies.

This project studies whether daily financial news headlines contain useful information about next-day movements in the S&P 500. In particular, we want to know whether headline sentiment is associated with short-run market direction and whether that relationship changes during different economic conditions. Instead of assuming the market behaves the same way across all years, we will compare several major periods: the 2008–2009 financial crisis, the 2020 COVID-19 shock, and the 2022–2023 rate-hike cycle.

Our main research questions are:

1. Is headline sentiment statistically related to next-day S&P 500 returns?
2. Do finance-specific sentiment methods perform better than general sentiment tools?
3. Does predictive performance differ across calm and high-volatility market regimes?

These questions are relevant to investors evaluating news-based strategies, researchers interested in market efficiency, and data scientists applying natural language processing to financial decision-making.

Dataset

Primary source: the public Kaggle file “S&P 500 with Financial News Headlines (2008–2024)” (dyutidasmahaptra; CC BY-SA 4.0). It contains 19,054 rows and three columns : date, headline, and close covering January 2, 2008 through March 4, 2024, across roughly 4,000 trading days with multiple headlines per day.

Limitations and design choices: the original news source is undocumented and likely biased toward major outlets; some headlines are clearly off-topic and must be filtered; only daily closes are available, so intraday reactions cannot be observed; and headline density rises sharply over time (under 700 per ten-month window in 2008–2012 versus over 4,000 in 2023–2024). We treat this concept drift in the source itself as both a limitation and an object of analysis rather than a nuisance to hide.

Planned Analytic Approach

- **EDA and data quality:** headline volume by year, off-topic filtering via keyword and entity heuristics and so on.
- **Sentiment extraction** We will compare multiple approaches to measuring headline sentiment
- **Topic modeling:** using like BERTopic or LDA to group headlines into themes (monetary policy, earnings, geopolitics, tech, banking stress) and test whether specific topics are more predictive than others.
- **Predictive modeling:** binary next-day direction and continuous next-day return, using logistic regression. All models use strict validation splits to avoid look-ahead bias.
- **Regime-based evaluation:** segment the timeline and evaluate each model within each regime. Honest reporting of where models fail (e.g., COVID) is a central deliverable, not a footnote.
- **Economic significance:** translate model signals into a simple long/flat backtest with transaction cost

Team Contributions

1 — **Data preparation lead:** cleaning pipeline, macro data integration, data dictionary, EDA summaries.
2 — **NLP lead** topic modeling, sentiment scoring methods
3 — **Modeling lead:** feature engineering, logistic regression and XGBoost, cross-validation design
4 — **Evaluation lead:** regime-segmented evaluation, backtest engine, robustness checks, model-failure case studies.
5 — **Writing and presentation lead:** report narrative, visualizations, presentation